

SARAJEVO, Bosnia and Herzegovina

WMO: 133530

Lat: 43.825N

Lon: 18.331E

Elev: 1708

StdP: 13.81

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.9	14.2	5.1	7.6	11.4	10.4	9.9	16.1	21.2	50.3	18.0	50.4	2.1	290	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.6	91.6	67.4	88.1	67.3	84.6	66.6	70.9	84.9	69.1	82.9	67.8	81.0	6.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	103.2	77.5	64.6	97.0	74.9	62.9	91.5	72.6	35.7	84.6	34.3	82.7	33.0	82.4	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	13.8	10.8	DB	0.7	97.2	8.3	3.6	-5.3	99.8	-10.2	101.9	-14.9	104.0	-20.9	106.6	(4)
(5)			WB	1.8	74.1	8.0	1.8	-3.9	75.4	-8.6	76.5	-13.1	77.5	-18.9	78.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	31.5	34.8	42.6	51.1	58.7	65.9	69.2	69.5	60.5	52.3	43.1	33.0		
	DBStd	15.25	8.84	8.90	7.87	6.52	5.96	6.38	5.56	5.41	6.33	7.17	8.59	8.68		
	HDD50	2155	576	428	247	62	7	1	0	0	3	62	238	531		
	HDD65	5521	1040	846	695	417	206	65	25	21	160	396	657	993		
	CDD50	2556	1	2	16	96	276	479	595	606	319	132	30	4		
	CDD65	447	0	0	0	0	10	93	155	162	26	1	0	0		
	CDH74	5406	0	0	1	42	274	1125	1783	1799	362	20	0	0		
	CDH80	2100	0	0	0	3	44	389	763	793	107	1	0	0		
(14) Wind	WSAvg	3.3	2.9	3.6	4.2	4.1	3.5	3.4	3.4	3.0	2.9	2.5	3.4	3.2		
(15) Precipitation	PrecAvg	41.3	2.9	3.0	3.0	3.3	3.3	3.7	3.2	2.8	4.0	4.1	4.3	3.8		
	PrecMax	59.4	7.0	7.2	6.3	7.0	7.4	8.5	5.8	7.0	10.4	8.5	8.5	10.6		
	PrecMin	29.9	0.6	0.6	0.2	0.6	1.7	0.5	0.8	0.3	0.9	0.3	0.7	0.0		
	PrecStd	7.8	1.6	1.7	1.6	1.5	1.4	1.8	1.4	1.7	2.4	2.0	1.8	2.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	62.6	69.9	79.0	84.4	91.5	96.6	96.9	88.2	76.8	68.0	59.1		
		MCWB	48.9	50.3	51.7	57.1	63.6	68.5	67.1	67.1	64.1	61.7	56.1	52.1		
	2%	DB	53.3	57.1	64.7	73.5	80.6	87.9	91.7	91.8	82.5	73.0	64.4	55.2		
		MCWB	47.0	47.3	50.0	55.9	62.5	67.9	67.7	66.9	63.9	60.3	55.2	48.6		
	5%	DB	48.2	53.2	61.0	69.5	77.0	84.5	88.0	88.0	77.4	68.4	61.0	50.3		
		MCWB	43.9	46.0	48.7	53.8	60.8	67.1	67.7	67.1	62.7	58.2	52.9	45.7		
	10%	DB	43.2	48.6	57.2	64.7	73.3	80.8	84.1	84.2	73.5	64.7	57.4	46.4		
		MCWB	41.0	43.3	47.4	52.1	59.6	65.6	66.9	66.5	61.2	56.9	51.0	44.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	51.1	53.8	59.4	66.4	72.6	72.9	73.0	67.7	63.8	58.0	52.7		
		MCDB	55.3	59.2	65.3	74.3	79.5	85.6	86.9	86.9	79.4	73.5	64.9	58.0		
	2%	WB	47.5	48.4	51.4	56.8	64.0	70.2	70.8	70.5	65.6	61.4	55.5	48.9		
		MCDB	52.6	56.2	62.0	69.8	76.3	84.0	85.1	84.5	78.2	69.6	63.0	53.7		
	5%	WB	44.2	46.2	49.7	55.0	62.1	68.1	69.0	68.7	63.9	59.3	53.6	46.8		
		MCDB	47.8	52.6	59.1	65.9	73.6	81.2	83.1	82.3	75.4	66.5	60.3	50.5		
	10%	WB	41.1	43.5	48.0	53.4	60.3	66.3	67.6	67.3	62.2	57.6	51.2	43.6		
		MCDB	43.4	48.1	56.9	62.8	70.9	78.8	80.9	80.4	71.5	63.7	57.0	46.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	16.6	19.8	21.9	23.2	23.3	24.6	25.9	22.0	20.5	15.9	12.0		
		MCDBR	19.7	24.2	29.3	31.9	31.3	30.3	32.5	34.1	30.4	27.6	21.9	17.5		
		MCWBR	13.8	15.6	16.2	15.7	15.1	13.6	12.8	12.9	13.8	15.8	13.7	12.6		
	5% WB	MCDBR	18.1	22.2	25.3	26.1	27.9	28.0	28.1	28.7	25.5	23.7	19.9	16.2		
		MCWBR	13.1	14.6	14.5	13.9	14.2	13.3	12.6	12.5	12.4	14.1	13.0	11.8		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.351	0.398	0.454	0.449	0.441	0.441	0.442	0.416	0.381	0.343	0.319		
	taud		2.373	2.252	2.175	2.059	2.123	2.203	2.203	2.197	2.245	2.334	2.390	2.390		
	Ebn at Noon		253	262	263	257	262	265	263	258	255	249	242	241		
	Edh at Noon		25	33	41	50	48	45	44	43	38	31	25	23		
(44) All-Sky Solar Radiation	RadAvg		456	679	1051	1347	1667	1912	1982	1785	1247	864	528	390		
	RadStd		64	82	141	175	149	163	152	168	122	97	65	69		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (29 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	-191	+202	N/A

Nomenclature: See separate page